

Performance Testing

Date	07 July 2026
Team ID	SWTID-2026-2168
Project Name	EduGenie: Google Gemini-Powered Learning Assistant
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Web Browser (Chrome)
Type of Testing	Functional & Basic Performance Testing
Target Module / API	AI Chat, Summary, Quiz, Flashcards
Test Environment	Local DevTunnel
Test Date	05 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Open Home Page	1	30 sec	Page loads successfully
2	AI Chat Request	1	60 sec	AI response generated
3	Generate Summary	1	45 sec	Summary displayed
4	Generate Quiz	1	60 sec	Quiz generated correctly

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	Observed during browser testing	Pass	Responsive
2	Response Time (Max)	< 5 seconds	Observed during browser testing	Pass	Stable
3	Throughput (Req/sec)	N/A	Not Measured	N/A	Browser testing only
4	Error Rate	< 1%	No major errors observed	Pass	Stable
5	CPU Utilization	< 80%	Not Measured	N/A	No profiling performed
6	Memory Utilization	< 80%	Not Measured	N/A	No profiling performed

Step 4: Observations & Analysis

Key Findings:

The application loaded successfully and all major features worked correctly during browser-based testing. All responses were generated without major issues.

Bottlenecks Identified:

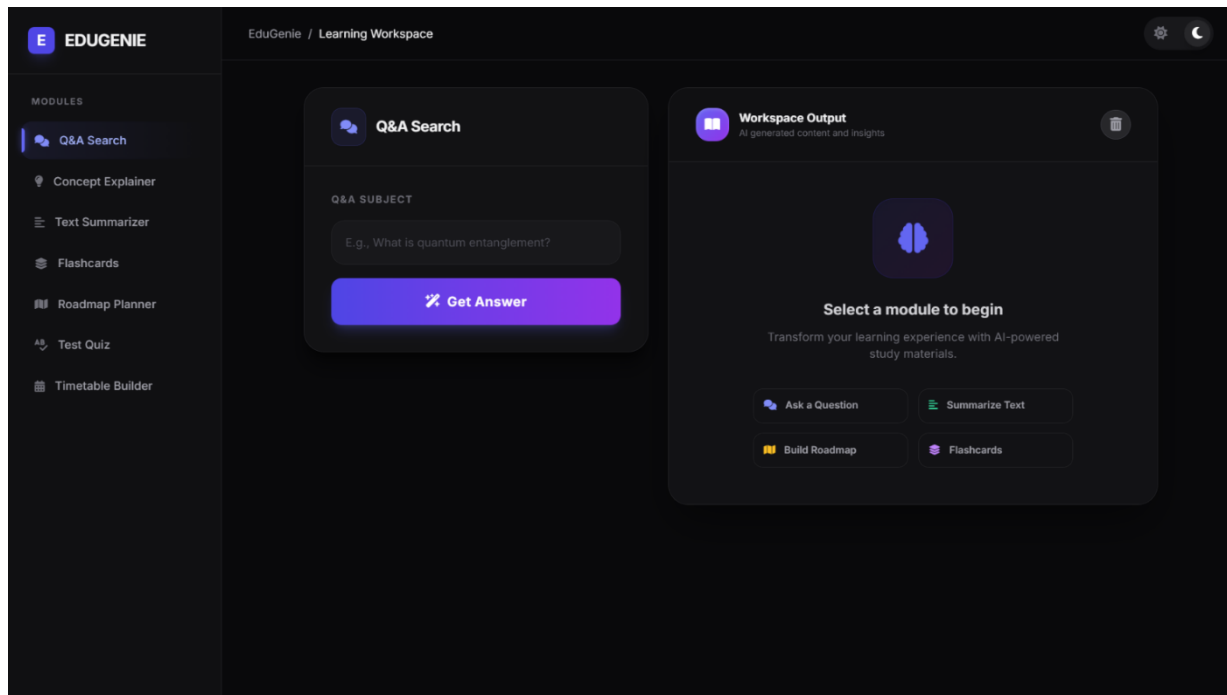
Occasional response delay due to Gemini API network latency.

Optimization Steps Taken:

Optimized frontend requests and improved backend API handling.

Step 5: Screenshots / Evidence

Screenshot 1: Home Page



Screenshot 2: AI Chat

